LAB=5

01 Stack using Array (Push, POP, Peek, Display)

```
#include<iostream>
Using namespace Std;
int st[5], top=-1;

Void push (int u) {
if (top==4)
 Cout<<" Overflow In";
else st[++top]=u; }
Void pop() {
if (top==-1)
Cout<<" underflow In";
 else
    Cout<<"Popped :"<<st[top--]
    <<"In"; }
Int peek() {
return top==-1 ? -1 : st[top]; }
Void display () {
for(int i=top; i>=0; i--)
Cout<<st[i]<<" ";
Cout<<" In";
```

```
int main() {
Push(10); Push(20); Push(30);
 display();
Cout << " Top " << Peek() << "\n";
Pop();
display();
}
```

Sample O/P :-

30 20 10

Top : 30

Popped : 30

20 10

02 Stack Overflow :-

```
#include <iostream>
Using namespace std;
int main() {
   int st[3], top=-1;

for(int i=1; i<=4; i++){
  if(top==2) cout<<" Stack Overflow!
    Cannot push "<<i<<"\n";
  else {
    st[++top]=i;

}
}
```

O/P :-

Stack Overflow! Cannot push 4

Q3 Stack Underflow :-

```
#include <iostream>
using namespace std;
int main() {
    int st[3], top = -1;
    if (top == -1) cout << "Stack underflow ! nothing to pop \n";
    else cout << st[top--];
}
```

O/P :-

Stack underflow ! nothing to pop

Display Top using peek :-

```
#include <iostream>
Using namespace std;
int main(){
  int St[5], top=-1;
  St[++top] = 10;
  St[++top] = 20;
  St[++top] = 30;
  Cout<<" Top element : "<<
    (top == -1 ? -1 : St[top])<<"\n";
}
```

O/p :-

#include <iostream>
Using namespace std;

int main() {
int st[100], top=-1, ch, x;
do {

cout<<"\n Push 2. pop 3, Peek 4. Displ
Exit :";

cin >> ch;
if (ch==1) {
if (top==99) cout<<" Overflow\n
else { cout<<"Value :";
cin>>u;
st[++top]=u; }

}

else if (ch==3) {
if (top==-1) cout<<" Empty\n";
else cout<<" Top : "<<st[top]<<"\n
} else if (ch==4) {
for(int i=top; i<=0; i--)
Cout<< st[i]<<" ";

```
        cout << "\n";
    }
    while (ch!=5);
}
```

:- Sample O/P :-

1. Push 2. Pop 3. peek 4. Display 5. Exit : 1

Value : 10

1. Push 2. pop 3. peek 4. Display 5. Exit : 1

Value 20

1. Push 2. pop 3. peek 4. Display 5. Exit : 4

20 10

Q6 Stack of Integers using Array :-

```
#include <iostream>
Using namespace Std;
Int main() {
  int St[100], top = -1, n, u;
  Cout << " How many Integers : ";
  Cin >> n;
  While( n-- ) { Cin >> u;
    St[++top] = u; }
  Cout << "popping : ";
  While (top>=0) Cout << St[top--]
    << " ";
}
```

Sample O/P :-

How many Integers : 3
10 20 30
Popping : 30 20 10

Q4 Reverse a String using Stack

```
#include <bits/stdc++.h>
Using namespace std;
int main(){
    String s;
    Stack <char> st;
    Cout << " Enter String : ";
    Cin >> s;
    for( char c : s) st.push(c);
    While( !st.empty()) { Cout << st.top();
                          st.pop(); }
}
```

O/P :-

Enter String : hello

olleh

Q8 Balanced Parenthese :-

```
#include<bits/stdc++.h>
using namespace std;
int main(){
    string s;
    stack<char> st;
    bool ok=true;
    cin>>s;
    for(char c:s){
        if(c=='('||c=='{'||c=='[')
            st.push(c);
        else{
            if(st.empty()){ ok=false;
            break;}
        char t=st.top();
        st.pop();
        if((c==')'&&t!='(')||
           (c=='}'&&t!='{')||
           (c==']'&&t!='['))
           {ok=false; break;}
    }}
    cout<<(ok&&st.empty()?
```

Infix to postfix :-

```
#include <bits/stdc++.h>
using namespace std;
int pr(char c){ return (c=='+'||c=='-')?1:(c=='*'||c=='/')?2:0;}
int main(){
    string s, r;
    stack<char> st;
    cin>>s;
    for(char c: s){
        if(isalnum(c)) r+=c;
        else if(c=='(') st.push(c);
        else if(c==')'){
            while(st.top()!='(') r+=st.top();
                st.pop();
            st.pop();
        }
        else{
            while(!st.empty()&&pr(st.top())>=pr(c)) r+=st.top();
            st.pop();
```

```
        st.push(c);
    }
    while(!st.empty()) H+=st.top
        st.pop();
    cout<<"Postfix: "<<H;
}
```

O/P:-

A+B+C-D

Postfix: ABC*-D

Evaluate postfix (Single-digit operands)

```
#include<iostream>
using namespace std;
int main(){
 string s;
 stack<int> st;
  cin>>s;
 for(char c:s){
    if(isdigit(c)) st.push(c-'0');
    else{
       int b = st.top(); st.pop();
       int a = st.top(); st.pop();

    st.push(c=='+'? a+b : c==
          '-'? a-b ; c=='*' a*b
          : a/b);

}
}
cout<< "Result" << st.top();
}
```

O/P:- 2 3 1 * + 9 -

Result : -4

```
#include<bits/stdc++.h>
using namespace std;

int pr(char c){return
(c=='+'||c=='-')?1:(c=='*'||
    c=='/')?2:0;
}int main(){
    string s,r;
    stack<char> st;
    cin>>s;
    reverse(s.begin(),s.end());
for(char &c:s)c=(c=='(')?')':
    (c==')')?'(':c;

for(char c:s){
    if(isalnum(c))r+=c;
    st.pop();
}else{
while(!st.empty()&&pr(st.top())
    >pr(c))r+=st.top().st.pop();
st.push(c);
}}
```

```
while ( !st.empty() ) {
    H += St.top(), St.pop();
reverse ( H.begin() , H.end() );
cout << "Prefix : " << H;
}
```

Sample O/p :-

A + B * C

Prefix : + A * B C

6B Stack using a class

```
#include <iostream>
Using namespace Std;
  Class Stack <
      int a[100], top=-1;
  Public:
      Void push(int x) <
      if (top == 99) Cout << "Overflow In";
      else a [++top ] = x; }
     Void pop() <
     if (top == -1)
      Cout << "underflow In";
      else
          Cout << " popped : " <<
      a [top--] << " In";
  >
     int peek() <
  return top == -1 ? -1 : a[top]; }
  Void display () < for (int i= top;
      i>= 0; i--) Cout << a[i] << " ";
```

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          Cout << " popped : " <<
      a[top--] << "\n";
     }
     int peek() {
    return top == -1 ? -1 : a[top]; }
    Void display() { for (int i= top;
        i>= 0; i--) Cout << a[i] << " ";
```

```
cout<<"\n"; }
};
int main(){
    Stack s;
    s.push(10);
    s.push(20);
    s.push(30);
    s.display();
    cout<<"Top:"<<s.peek()<<"\n";
    s.pop();
    s.display();
}
```

O/P:-

30 20 10

Top : 30

Popped : 30

20 10

Q.4 Stack Us in Recursion

```
#include <iostream>
using namespace std;
int fact(int n){
  if(n<=1){cout<<"fact(1)
  return 1\n";
  return 1;
}
  cout<<"fact("<<n<<") called\n";
  int H=n*fact(n-1);
  cout<<"fact("<<n<<") returns "
  <<H<<"\n";
  return H;
}
int min(){fact(4);}
```

Sample O/P 1-

fact(4) called
"
"
fact(1) return 1
fact(2) return 2
fact(3) return 6
fact(4) return 24

Time Complexity of Push, Pop, Peek, Display

```
#include <iostream>
Using namespace std;
Using namespace chrono;

int main() {
    const int n = 1000000;
    Vector< int > St;
    St.reverse(N);
    auto t = steady_clock::now();
    auto lap = [&](string name) {
      auto now = steady_clock::now();
    cout << name << ": " << duration_cast
        <microseconds(now - t).count()
          << us \n";
        t = now;
};
for( int i=0; i<n; i++)
    St.pust_back(i);
    lap("Push  (N times) ");
    for( int i=0; i<n; i++)
      Volatile int n = St.back();
      lap("Peek  (N times) ");
```

```
long long Sum = 0;
for ( int i = n-1; i >= 0; i--)
    Sum += St[i];
while (!st.empty())
    st.pop_back();
```

Sample O/p :-

Push (n times) ; 2500 us

Peek (n times) ; 800 us

Display (once) ; 700 us

Pop (n times) ; 600 us